

Climate Alpha: The Hypothesis We Tested and Rejected

Supporting research note for the MFE 230GA final project

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Summary. After Homework 1, each of us worked with ChatGPT and Claude to test whether emissions data could support a tradable Climate Alpha strategy. That work is in two repositories, 230GA-Final-Project-main and 230GA-Final-Project-hashim-research-extension. The idea was economically sensible and passed an initial smell test, so we built it properly and stress-tested it. It did not survive: no version gave a robust, implementable alpha. We chose not to force a result. We recorded the idea as falsified and moved to a different source of alpha, which is analysed in the main final report. We report the Climate work here because it explains our research path and the evidence that led to the pivot.

1 Research motivation and initial thesis

Homework 1 asked us to compare high- and low-emission Fama–French industries and to use an AI assistant to propose strategies. Our group selected a factor-neutral Green-minus-Brown spread (the five lowest- minus the five highest-intensity industries) and wrote down the risk in advance: if the hedged alpha came out at zero, the spread would be “a growth-minus-value bet with a commodity short attached, not a climate factor.”

The literature gives two reasons a climate return could exist. Investors with green preferences may accept lower expected returns on green assets, so brown firms carry a risk or taste premium (Bolton and Kacperczyk, 2021). But realised green returns can also be high simply because climate concern rose unexpectedly: in the model of Pástor et al. (2021), green stocks outperform when concern strengthens by surprise, and Pástor et al. (2022) attribute most of the 2012–2020 green outperformance to such shocks. That second channel suggested a timing strategy rather than a permanent tilt. Our thesis, stated so it could fail, was:

The Green-minus-Brown spread has no stable premium, but its factor-neutral Brown leg underperforms in the months after climate-transition attention spikes above its own historical range. If a holdout window never used in design shows no such profit, the thesis is wrong.

2 What we tried

The full program has more than twenty analyses. Five of them explain how our view changed.

(1) The team’s timing rule (main repository). Industry returns are Ken French FF49 value-weighted portfolios; emissions intensity is the Homework 1 course file (one undated cross-section, 41 of 49 industries). The Brown leg (Util, Ships, Aero, Steel, BldMt) is hedged with rolling 60-month FF3 betas lagged one month, as Homework 1 proposed. When the 60-month z-score of the attention series crosses its past-only 80th percentile, the rule shorts the hedged Brown leg for 3 or 6 months, sized to 5% residual volatility. A “Pure” version first removes the part of attention explained by lagged

macro data, and a continuous version scales the position smoothly with the signal. Costs are charged on the leg and on every overlay. Design used 2010-01 to 2022-07; 2022-08 to 2026-07 was kept as an untouched holdout.

(2) A signal audit (extension, module M1). We checked what the “attention” column actually is, and fed the same rules two true climate-concern indices: media climate change concern (MCCC, Ardia et al., 2023) and climate policy uncertainty (CPU, Gavrilidis, 2021).

(3) Timing versus exposure. Every timing claim was compared with an always-short position: the same hedged Brown leg, hedge and volatility target, with no signal.

(4) A frozen out-of-sample test (M8). A cleaner signal (the regulation share of volatility news, zeros treated as missing) and a four-part pass bar were written down and hashed before any 1994–2009 return was used, then run once.

(5) Alternatives on the same data (M4, M5, M7). We rebuilt the emissions ranking for all 49 industries from EPA supply-chain factors, tried carbon-aware industry momentum, and put every test from every module into one multiple-testing ledger.

Two input problems were fixed before the new tests and are used in Table 2 (“corrected baseline”): attention is lagged one month, because we could not confirm it is published by month-end, and the missing October 2025 CPI print is interpolated. Neither changes a verdict. AI coding agents wrote most of the extension’s code under Hashim’s direction, and a second, independent implementation reproduced each module’s headline numbers before we used them.

3 What we found

Table 1 condenses the results. Table 2 and Figure 1 show the time pattern.

Table 1: The experiments that changed our view. Alphas are annualised; t -statistics are Newey–West with 6 lags. Rows 1 and 3 are reproduced in `Climate_Alpha_Research_Summary.ipynb`; the other rows are verified extension results copied with their source file into `data/verified_extension_results.csv`.

Experiment	What looked promising	What happened
Team rules as first built (validation 2010-2022 vs. holdout 2022-2026)	Original 6m validation net 2.45%, FF3 alpha $t = 2.23$	every discrete rule lost 2.1% to 3.8% a year in the holdout
What the attention series measures	label: climate-transition attention	equals FRED EMV energy/environmental-regulation volatility tracker in 500/500 months; corr. with MCCC climate concern -0.0004 ; 0 of 12 climate-index validation alphas with $t \geq 1.96$
Timing vs. simply being short Brown (COVID)	Original 3m COVID FF3 alpha 6.38% ($t = 3.62$)	always-short alpha 4.54% ($t = 1.96$); paired edge 1.84% ($t = 0.87$)
Frozen pre-registered rule, 1994-2009	a cleaner signal (regulation share of volatility news), frozen before testing	timing alpha -0.18% a year ($t = -0.27$); calendar-shuffle $p = 0.48$; fails 3 of 4 bars
Alternatives on the same data	EPA supply-chain ranking; post-2010 FF5+UMD alpha 4.7%; carbon-aware industry momentum	EPA alpha $t = 1.52$; momentum book FF5+UMD alpha $t = 1.09$ with UMD beta 0.99
Multiple testing across the whole program	several individual $t > 2$ results	73 module primary alphas: smallest Holm-adjusted $p = 1.000$

Promising in sample. In the design window the 6-month rule earned 2.45% a year net with an FF3 alpha t of 2.23, and the 3-month rule earned an alpha of 6.38% ($t = 3.62$) during COVID.

Weaker once the right benchmark is used. The raw spread loads -0.23 on HML over 1970–2026 ($t = -4.63$) and -0.30 since 2010 ($t = -5.74$), with no significant alpha. The hedge removes most of that exposure, but the COVID gain is mostly the exposure itself: the always-short position earned 4.54% ($t = 1.96$), and the rule’s paired edge over it was 1.84% a year ($t = 0.87$).

Failed out of sample. Over the untouched holdout every discrete rule lost money (Table 2), and the signal’s IC flipped sign: for the continuous weight it went from -0.15 ($t = -2.19$) in design to $+0.06$ in the holdout. The frozen 1994–2009 rule had a timing alpha of -0.18% a year ($t = -0.27$), and 5,000 random re-datings of its positions gave a one-sided p of 0.48.

Implementation and data problems. The “climate-transition attention” column equals FRED series EMVENRGYENVREG in all 500 months: the Equity Market Volatility tracker for energy and environmental regulation (Baker et al., 2019), a count of articles about stock-market volatility. Its z-score is uncorrelated with MCCC ($r = -0.0004$), and it is exactly zero in 39 of the 59 months since October 2021. The emissions file is one undated snapshot applied back in time, a look-ahead in leg membership; the EPA rebuild agrees with it broadly but places Ships and Aero, two of the five Brown industries, near mid-table.

Costs were not the binding problem. The discrete rules turn over 2.7 to 5.3 times a year and costs take at most 0.6 percentage points a year (2010–2026), but their holdout FF3 alphas are already negative before costs (at best -1.5% a year).

Insights that survived. Each apparent alpha in the program was a known factor in disguise: value in the raw spread, the Brown-leg exposure in COVID, and UMD in industry momentum (equal-weight momentum book: FF5+UMD alpha $t = 1.09$, UMD beta 0.99). We also learned to read the label of every input and to benchmark every timing rule against its untimed version.

Table 2: Time pattern, corrected baseline. Each cell: annual net return / FF3 alpha (NW(6) t), in %. Recent windows have 12–18 observations and are low power.

Window	Original 3m	Original 6m	Always-short Brown
Validation (2010-01–2022-07), $n=151$	1.78 / 1.62 (1.97)	2.16 / 2.09 (2.18)	1.71 / 1.39 (1.12)
Holdout (2022-08–2026-07), $n=48$	$-2.57 / -2.16 (-1.90)$	$-1.84 / -1.97 (-1.30)$	$-1.04 / -1.72 (-1.09)$
COVID (2020-01–2021-12), $n=24$	8.04 / 6.38 (3.62)	5.95 / 3.99 (1.79)	6.87 / 4.54 (1.96)
Last 18m (2025-02–2026-07), $n=18$	$-4.51 / -4.05 (-1.55)$	$-1.35 / -4.06 (-1.80)$	$-1.07 / -6.25 (-1.87)$

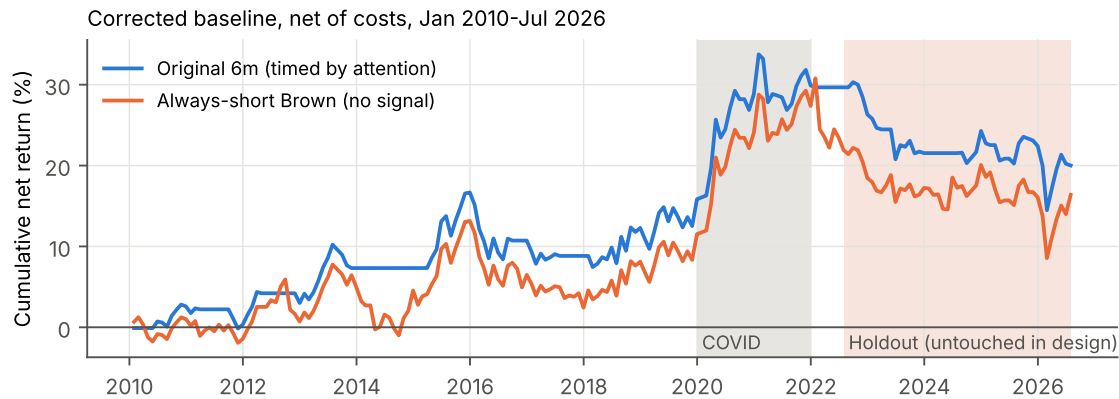


Figure 1: Cumulative net return of the 6-month timing rule and the untimed always-short position, January 2010 to July 2026. The two lines move together; the timing rule adds little beyond the exposure, and both lose in the holdout.

4 Why we did not implement the strategy

We ranked the reasons by how much each one, on its own, would block implementation.

1. **Out-of-sample failure.** The holdout and the frozen pre-2010 test both failed, and the signal's IC reversed sign out of sample. This alone rules out capital.
2. **The signal does not measure what the thesis needs.** It is volatility news, not climate concern, and genuine climate-concern indices earn nothing through the same rules (0 of 12 validation alphas with $t \geq 1.96$).
3. **Alpha cannot be separated from known exposures.** The best result is mostly an untimed short of the Brown leg; the raw spread is a short-value bet.
4. **Multiple testing.** Across 73 module primary alpha tests the smallest Holm-adjusted p is 1.00. Results that looked convincing alone do not survive the number of things we tried.
5. **Data limits.** A single, undated emissions snapshot and a signal that is mostly zero after 2021 make any live version fragile.

5 The pivot

Because the Climate Alpha hypothesis did not survive the robustness we required, we did not implement it, and we did not search for a specification that would make it work. The work did leave us with one clear lesson: a strategy is only interesting if it survives the factor that most resembles it. Our industry-momentum checks showed that the relevant factor for industry signals is UMD. That led directly to the main project: *do analyst EPS revisions carry information about industry returns beyond industry momentum, once FF5 and UMD are controlled for?* The main report tests that question with a falsification criterion fixed in advance.

References

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